First, this story illustrates a mode of astronomical discovery quite different from that illustrated in Chapter 8. In Chapter 8, Zel'dovich and Novikov proposed a concrete method to search for black holes; experimental physicists, astronomers, and astrophysicists implemented that method; and it paid off. In this chapter, gigantic black holes are already being observed by Reber in 1939, long before anyone ever thought to look for them, but it will take forty years for the mounting observational evidence to force astronomers to the conclusion that black holes are what they are seeing.

Second, Chapter 8 illustrated the powers of astrophysicists and relativists; this chapter shows their limitations. The types of black holes discovered in Chapter 8 were predicted to exist a quarter century before anyone ever went searching for them. They were the Oppenheimer–Snyder holes: a few times heavier than the Sun and created by the implosion of heavy stars. The gigantic black holes of this chapter, by contrast, were never predicted to exist by any theorist. They are thousands or millions of times heavier than any star that any astronomer has ever seen in the sky, so they cannot possibly be created by the implosion of such stars. Any theorist predicting these gigantic holes would have tarnished his or her scientific reputation. The discovery of these holes was serendipity in its purest form.

Third, this chapter's story of discovery will illustrate, even more clearly than Chapter 8, the complex interactions and interdependencies of four communities of scientists: relativists, astrophysicists, astronomers, and experimental physicists.

Fourth, it will turn out, late in this chapter, that the spin and the rotational energy of gigantic black holes play central roles in explaining the observed radio waves. By contrast, a hole's spin was of no importance for the observed properties of the modest-sized holes in Chapter 8.

In 1940, having made his first radio scans of the sky, Reber carefully wrote up a technical description of his telescope, his measurements, and his map, and mailed it to Subrahmanyan Chandrasekhar, who was now the editor of the *Astrophysical Journal* at the University of Chicago's Yerkes Observatory, on the shore of Lake Geneva in Wisconsin. Chandrasekhar circulated Reber's remarkable manuscript among the Yerkes astronomers. Bemused by the manuscript and skeptical of this completely unknown amateur, several of the astronomers drove down to Wheaton, Illinois, to